Year 11 Foundation

REVISION PACK

Graphs 6

Name: _____

Class: _____

Graphs 6 · Quadratic graphs (Lesson 8.1)

Worked example: a table, a graph and solving

This question is about the graph of $y = x^2 - x - 2$.

x	−2	−1	0	1	2	3
y	4		−2			4

(a) Complete the table: find y when $x = -1$

(b) Draw the graph.

(c) Use your graph to solve $x^2 - x - 2 = 0$

(d) Use your graph to find estimates for the solutions of $x^2 - x - 2 = 4$

(a)	**Step 1**	Put $x = -1$ in	$(-1)^2 - (-1) - 2$
	Step 2	Square first, then add	$1 + 1 - 2 = 0$
	Step 3	Answer	$y = 0$
(b)	**Step 1**	Plot the points, join with a smooth curve	
(c)	**Step 1**	Read where it crosses the x-axis	
	Step 2	Answer	$x = -1$ or $x = 2$
(d)	**Step 1**	Draw the line	$y = 4$
	Step 2	Read where they cross	
	Step 3	Answer	$x = -2$ or $x = 3$

Practice: a table, a graph and solving

1.1 This question is about the graph of $y = x^2 - 4$

x	−3	−2	−1	0	1	2	3
y	5			−4			5

(a) Complete the table of values.

(b) Draw the graph for x from −3 to 3.

(c) Use your graph to solve $x^2 - 4 = 0$

(d) Use your graph to find estimates for the solutions of $x^2 - 4 = 1$

1.2 This question is about the graph of $y = x^2 + 2x - 3$

x	−4	−3	−2	−1	0	1	2
y	5			−4			5

(a) Complete the table of values.

(b) Draw the graph for x from −4 to 2.

(c) Use your graph to solve $x^2 + 2x - 3 = 0$

(d) Use your graph to find estimates for the solutions of $x^2 + 2x - 3 = 2$

Practice: a table, a graph and solving (continued)

1.3 This question is about the graph of $y = x^2 - 4x + 3$

x	−1	0	1	2	3	4	5
y	8			−1			8

(a) Complete the table of values.

(b) Draw the graph for x from −1 to 5.

(c) Use your graph to solve $x^2 - 4x + 3 = 0$

(d) Use your graph to find estimates for the solutions of $x^2 - 4x + 3 = 1$

1.4 This question is about the graph of $y = x^2 - 2x - 8$

x	−3	−2	−1	0	1	2	3	4	5
y	7				−9				7

(a) Complete the table of values.

(b) Draw the graph for x from −3 to 5.

(c) Use your graph to solve $x^2 - 2x - 8 = 0$

(d) Use your graph to find estimates for the solutions of $x^2 - 2x - 8 = -4$

1.5 This question is about the graph of $y = x^2 + x - 6$

x	−4	−3	−2	−1	0	1	2	3
y	6				−6			6

(a) Complete the table of values.

(b) Draw the graph for x from −4 to 3.

(c) Use your graph to solve $x^2 + x - 6 = 0$

(d) Use your graph to find estimates for the solutions of $x^2 + x - 6 = 2$

1.6 This question is about the graph of $y = x^2 - 4x$

x	−1	0	1	2	3	4	5
y	5			−4			5

(a) Complete the table of values.

(b) Draw the graph for x from −1 to 5.

(c) Use your graph to solve $x^2 - 4x = 0$

(d) Use your graph to find estimates for the solutions of $x^2 - 4x = 3$

Graphs 6 · Graphs to solve equations (Lesson 8.2)

Worked example: where a line meets a curve

The graphs of $y = x^2 - 2$ and $y = -x$ are drawn on a grid.

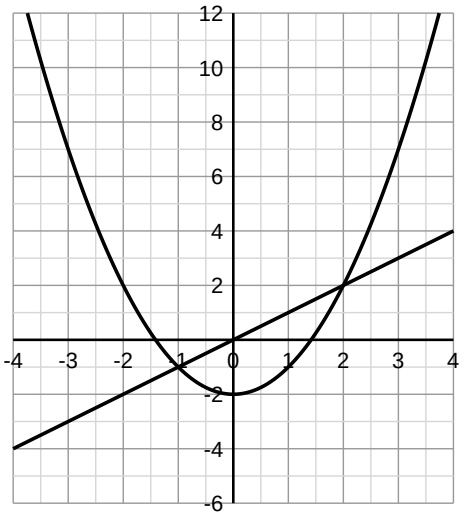
(a) Write down the coordinates of the points where the graphs intersect.

(b) The x-coordinates solve $x^2 + bx + c = 0$. Find b and c.

(a)	Step 1	Read where the graphs cross	
	Step 2	Answer	$(-2, 2)$ and $(1, -1)$
(b)	Step 1	Make them equal	$x^2 - 2 = -x$
	Step 2	Make one side 0	$x^2 + x - 2 = 0$
	Step 3	Answer	$b = 1,\ c = -2$

Practice: where a line meets a curve

2.1

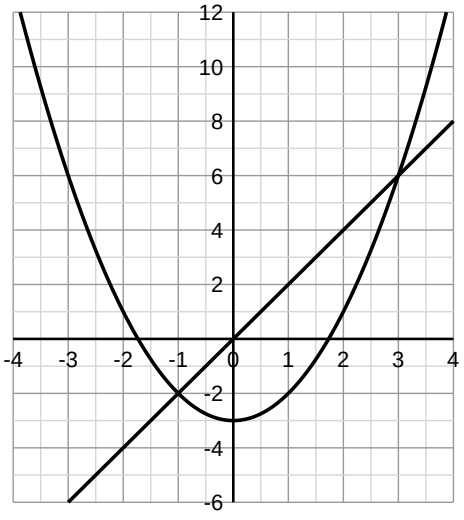


The graphs of $y = x^2 - 2$ and $y = x$ are drawn on the grid.

(a) Write down the coordinates of the points where the graphs intersect.

(b) The x-coordinates of these points are the solutions of an equation of the form $x^2 + bx + c = 0$. Find the values of b and c.

2.2



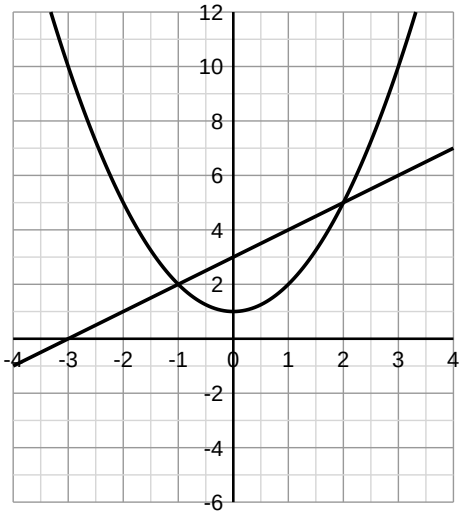
The graphs of $y = x^2 - 3$ and $y = 2x$ are drawn on the grid.

(a) Write down the coordinates of the points where the graphs intersect.

(b) The x-coordinates of these points are the solutions of an equation of the form $x^2 + bx + c = 0$. Find the values of b and c.

Practice: where a line meets a curve (continued)

2.3

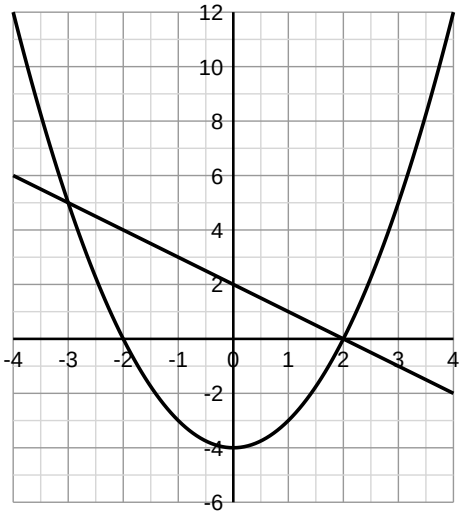


The graphs of $y = x^2 + 1$ and $y = x + 3$ are drawn on the grid.

(a) Write down the coordinates of the points where the graphs intersect.

(b) The x-coordinates of these points are the solutions of an equation of the form $x^2 + bx + c = 0$. Find the values of b and c.

2.4



The graphs of $y = x^2 - 4$ and $y = -x + 2$ are drawn on the grid.

(a) Write down the coordinates of the points where the graphs intersect.

(b) The x-coordinates of these points are the solutions of an equation of the form $x^2 + bx + c = 0$. Find the values of b and c.

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

Worked example: a line and the curve y = x²

Solve the simultaneous equations $y = x^2$ and $y = x + 20$

Step 1	Make them equal	$x^2 = x + 20$
Step 2	Make one side 0	$x^2 - x - 20 = 0$
Step 3	Factorise	$(x - 5)(x + 4) = 0$
Step 4	Each bracket = 0	$x = 5$ or $x = -4$
Step 5	Find y	$y = 5 + 20 = 25, \quad y = -4 + 20 = 16$
Step 6	Answer	$x = 5, y = 25$ or $x = -4, y = 16$

Practice: a line and the curve y = x²

3.1 Solve the simultaneous equations

$y = x^2$

$y = x + 2$

3.2 Solve the simultaneous equations

$y = x^2$

$y = 2x + 3$

3.3 Solve the simultaneous equations

$y = x^2$

$y = x + 12$

3.4 Solve the simultaneous equations

$y = x^2$

$y = 2x + 8$

3.5 Solve the simultaneous equations

$y = x^2$

$y = -x + 6$

3.6 Solve the simultaneous equations

$y = x^2$

$y = 4x - 3$

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

Worked example: a line and a curve with a number

Solve the simultaneous equations $y = x^2 - 1$ and $y = 3x + 9$

Step 1	Make them equal	$x^2 - 1 = 3x + 9$
Step 2	Make one side 0	$x^2 - 3x - 10 = 0$
Step 3	Factorise	$(x - 5)(x + 2) = 0$
Step 4	Each bracket = 0	$x = 5$ or $x = -2$
Step 5	Find y	$y = 3 \times 5 + 9 = 24, \quad y = 3 \times (-2) + 9 = 3$
Step 6	Answer	$x = 5, y = 24$ or $x = -2, y = 3$

Practice: a line and a curve with a number

3.7 Solve the simultaneous equations

$y = x^2 + 3$

$y = 3x + 1$

3.8 Solve the simultaneous equations

$y = x^2 - 2$

$y = x + 4$

3.9 Solve the simultaneous equations

$y = x^2 + 5$

$y = 4x + 2$

3.10 Solve the simultaneous equations

$y = x^2 - 3$

$y = 2x$

3.11 Solve the simultaneous equations

$y = x^2 + 4$

$y = -x + 10$

3.12 Solve the simultaneous equations

$y = x^2 - 1$

$y = -2x + 7$

Graphs 6 · Graphs to solve equations (Lesson 8.2)

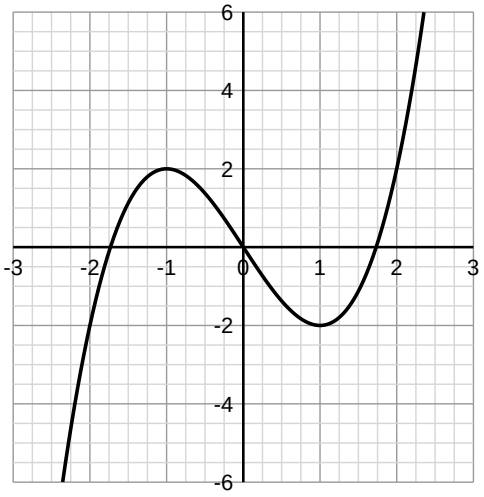
Worked example: solving with a cubic graph

The graph of $y = x^3 - 6x$ is drawn on a grid. Use the graph to find estimates for the solutions of $x^3 - 6x = 0$

Step 1	Read where it crosses the x-axis	
Step 2	Answer	$x = -2.4,\ 0,\ 2.4$

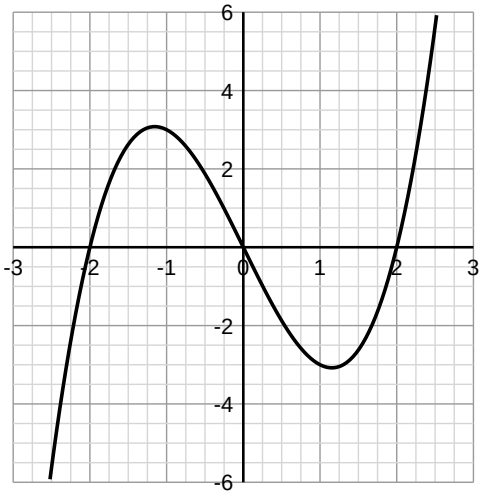
Practice: solving with a cubic graph

4.1



The graph of $y = x^3 - 3x$ is drawn on the grid.
Use the graph to find estimates for the solutions of $x^3 - 3x = 0$

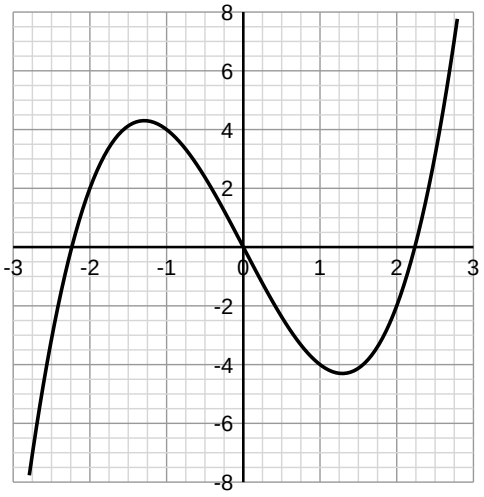
4.2



The graph of $y = x^3 - 4x$ is drawn on the grid.
Use the graph to find estimates for the solutions of $x^3 - 4x = 0$

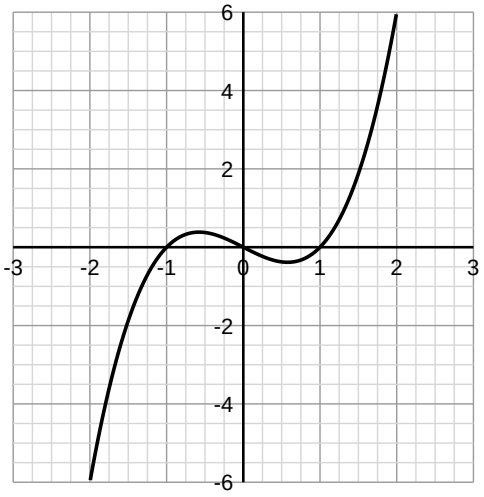
Practice: solving with a cubic graph (continued)

4.3



The graph of $y = x^3 - 5x$ is drawn on the grid.
Use the graph to find estimates for the solutions of $x^3 - 5x = 0$

4.4



The graph of $y = x^3 - x$ is drawn on the grid.
Use the graph to find estimates for the solutions of $x^3 - x = 0$

Worked example: turning points, roots and the y-intercept

The graph of $y = x^2 - 8x + 12$ is drawn on a grid.

(a) Write down the coordinates of the turning point.

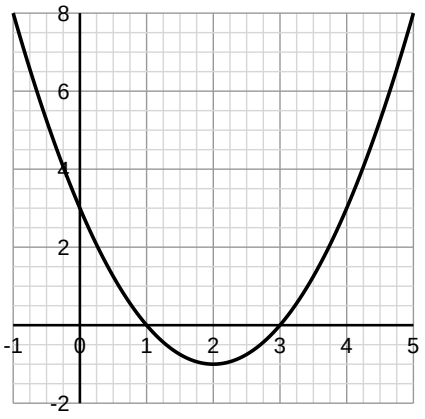
(b) Write down the solutions of $x^2 - 8x + 12 = 0$

(c) Write down where the graph crosses the y-axis.

(a)	Step 1	Find the lowest point	
	Step 2	Answer	$(4, -4)$
(b)	Step 1	Read where it crosses the x-axis	
	Step 2	Answer	$x = 2$ or $x = 6$
(c)	Step 1	Read where it crosses the y-axis	
	Step 2	Answer	$(0, 12)$

Practice: turning points, roots and the y-intercept

5.1



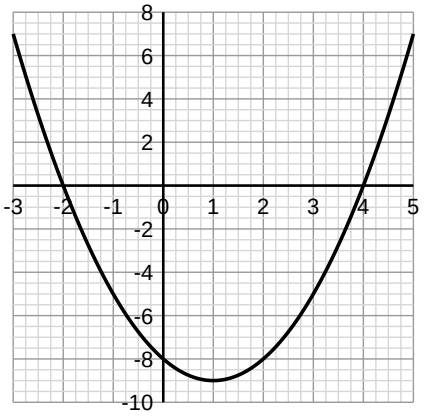
The graph of $y = x^2 - 4x + 3$ is drawn on the grid.

(a) Write down the coordinates of the turning point.

(b) Write down the solutions of $x^2 - 4x + 3 = 0$

(c) Write down the coordinates of the point where the graph crosses the y-axis.

5.2



The graph of $y = x^2 - 2x - 8$ is drawn on the grid.

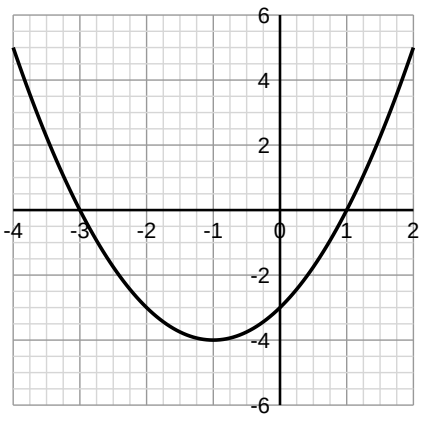
(a) Write down the coordinates of the turning point.

(b) Write down the solutions of $x^2 - 2x - 8 = 0$

(c) Write down the coordinates of the point where the graph crosses the y-axis.

Practice: turning points, roots and the y-intercept (continued)

5.3



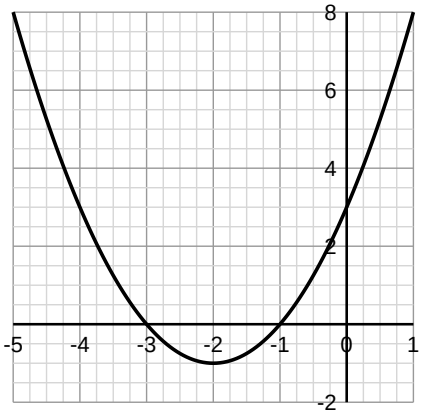
The graph of $y = x^2 + 2x - 3$ is drawn on the grid.

(a) Write down the coordinates of the turning point.

(b) Write down the solutions of $x^2 + 2x - 3 = 0$

(c) Write down the coordinates of the point where the graph crosses the y-axis.

5.4



The graph of $y = x^2 + 4x + 3$ is drawn on the grid.

(a) Write down the coordinates of the turning point.

(b) Write down the solutions of $x^2 + 4x + 3 = 0$

(c) Write down the coordinates of the point where the graph crosses the y-axis.

Graphs 6 · Quadratic graphs (Lesson 8.1)

Worked example: substituting into a quadratic

$y = x^2 - 6$

(a) Work out the value of y when $x = -4$ (b) Work out the positive value of x when $y = 30$

(a)	**Step 1**	Put $x = -4$ in	$y = (-4)^2 - 6$
	Step 2	Square first	$16 - 6 = 10$
	Step 3	Answer	$y = 10$
(b)	**Step 1**	Make it equal to 30	$x^2 - 6 = 30$
	Step 2	Add 6	$x^2 = 36$
	Step 3	Square root	$x = 6$
	Step 4	Answer	$x = 6$

Practice: substituting into a quadratic

5.5 $y = x^2 + 3$

(a) Work out the value of y when $x = -2$ (b) Work out the positive value of x when $y = 12$

5.6 $y = x^2 - 7$

(a) Work out the value of y when $x = -4$ (b) Work out the positive value of x when $y = 29$

5.7 $y = 2x^2 - 1$

(a) Work out the value of y when $x = -3$ (b) Work out the positive value of x when $y = 49$

5.8 $y = x^2 - 2$

(a) Work out the value of y when $x = -5$ (b) Work out the positive value of x when $y = 47$

5.9 $y = x^2 + 4$

(a) Work out the value of y when $x = -1$ (b) Work out the positive value of x when $y = 40$

5.10 $y = 3x^2 + 1$

(a) Work out the value of y when $x = -2$ (b) Work out the positive value of x when $y = 28$